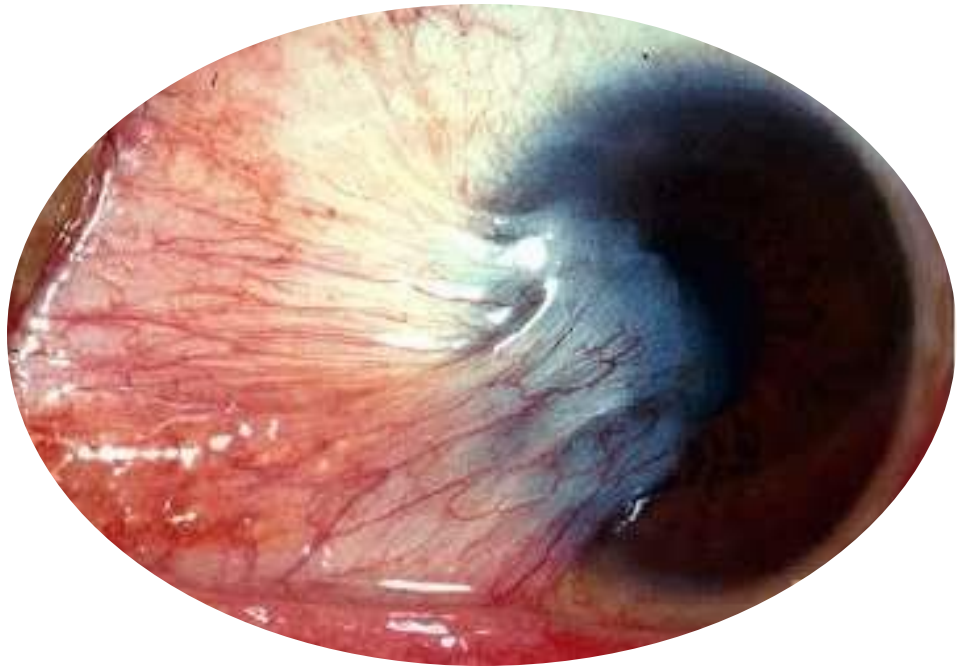
PTERYGIUM

WING SHAPED FOLD

SURFER'S EYE



PTERYGIUM

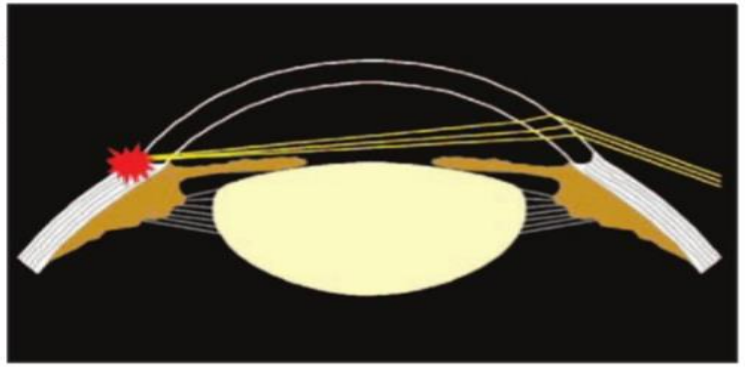
DR. RAMYA R NADIG

FINAL YEAR POST GRADUATE

OPHTHALMOLOGY

INTRODUCTION

- Greek word – “pteryx” meaning wing
- Definition: benign triangular fibrovascular proliferation presenting as superficial wing- shaped lesion extending from the perilimbal bulbar conjunctiva onto the corneal surface
- Nasal side
- Prevalent in – Pterygium belt … also Eskimos
- India – prevalence is 9.5 to 13 %



PSEUDOPTERYGIUM

- Fold of conjunctiva attached to cornea following injury to cornea
- Inflammatory process
- Any age, any site
- Stationary
- Probe can be passed beneath

GRADING SYSTEMS – 2

- MORPHOLOGICAL – Tan et al
  - T1 - ATROPHIC
  - T2 - INTERMEDIATE
  - T3 - FLESHY
- BASED ON THE EXTENT – Chao et al
  - GRADE 1- JUST CROSSING LIMBUS
  - GRADE 2- BTW LIMBUS AND PUPIL
  - GRADE 3- CROSSING PUPIL



Fig. 20.3: Clinical photograph depicting atrophic type of pterygium of right eye.



Fig. 20.4: Clinical photograph depicting an intermediate type of pterygium in the left eye.

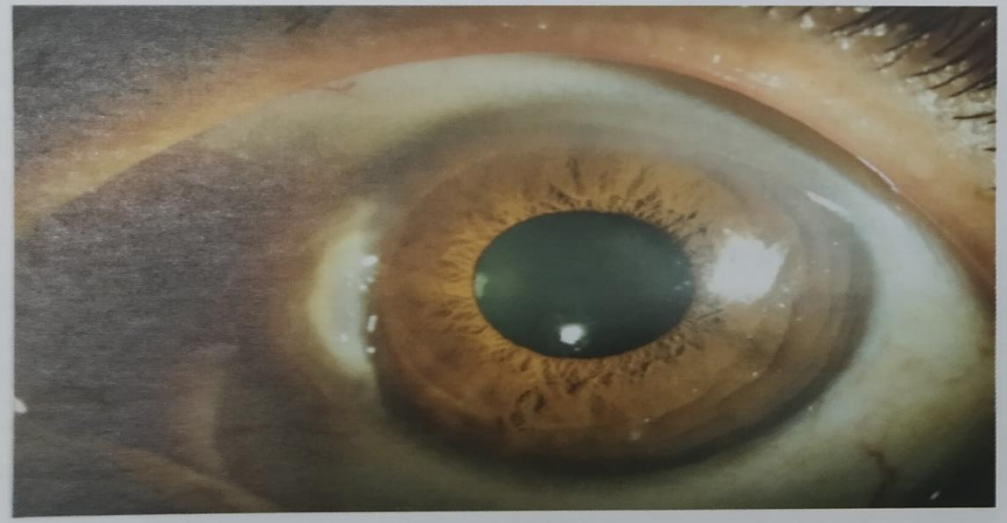


Fig. 20.5: Clinical photograph depicting a fleshy pterygium of left eye.

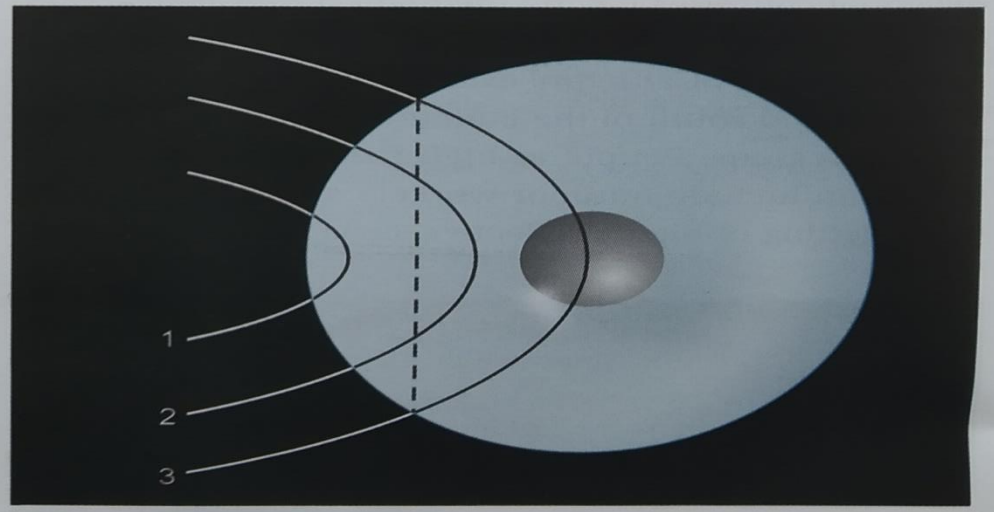
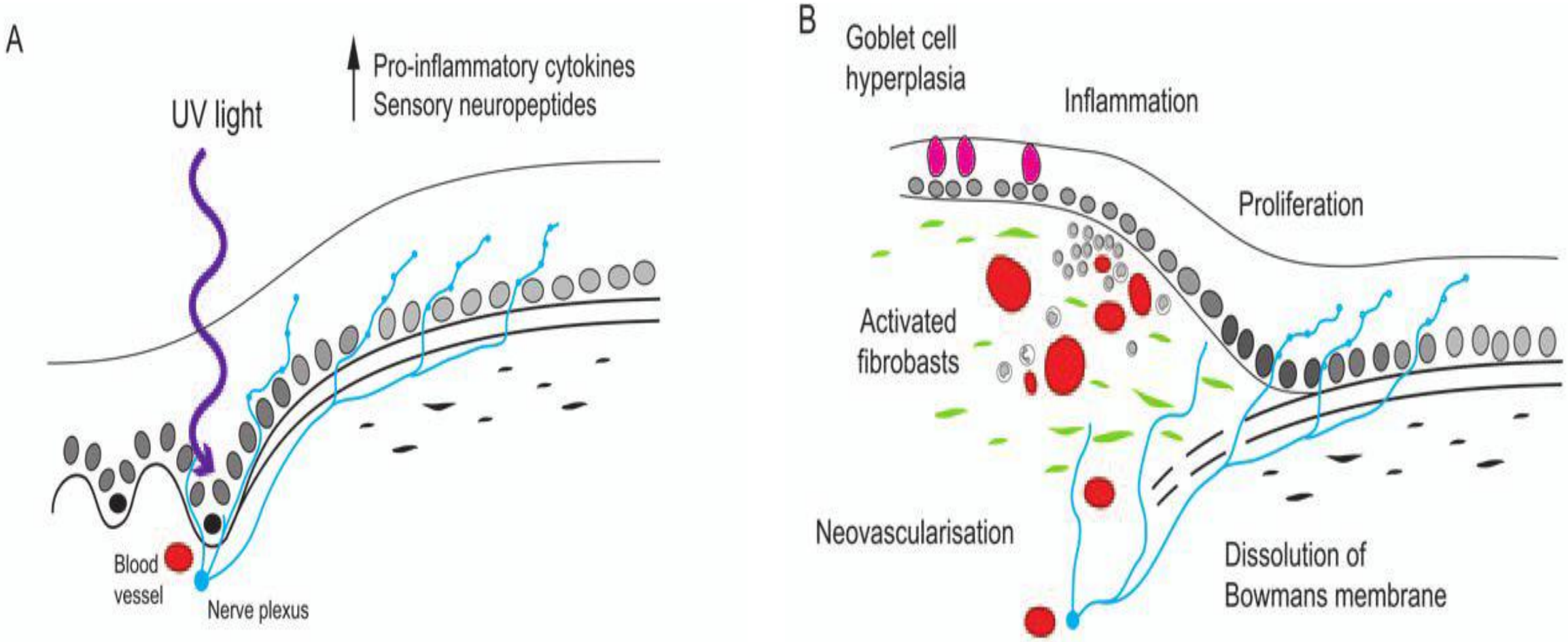


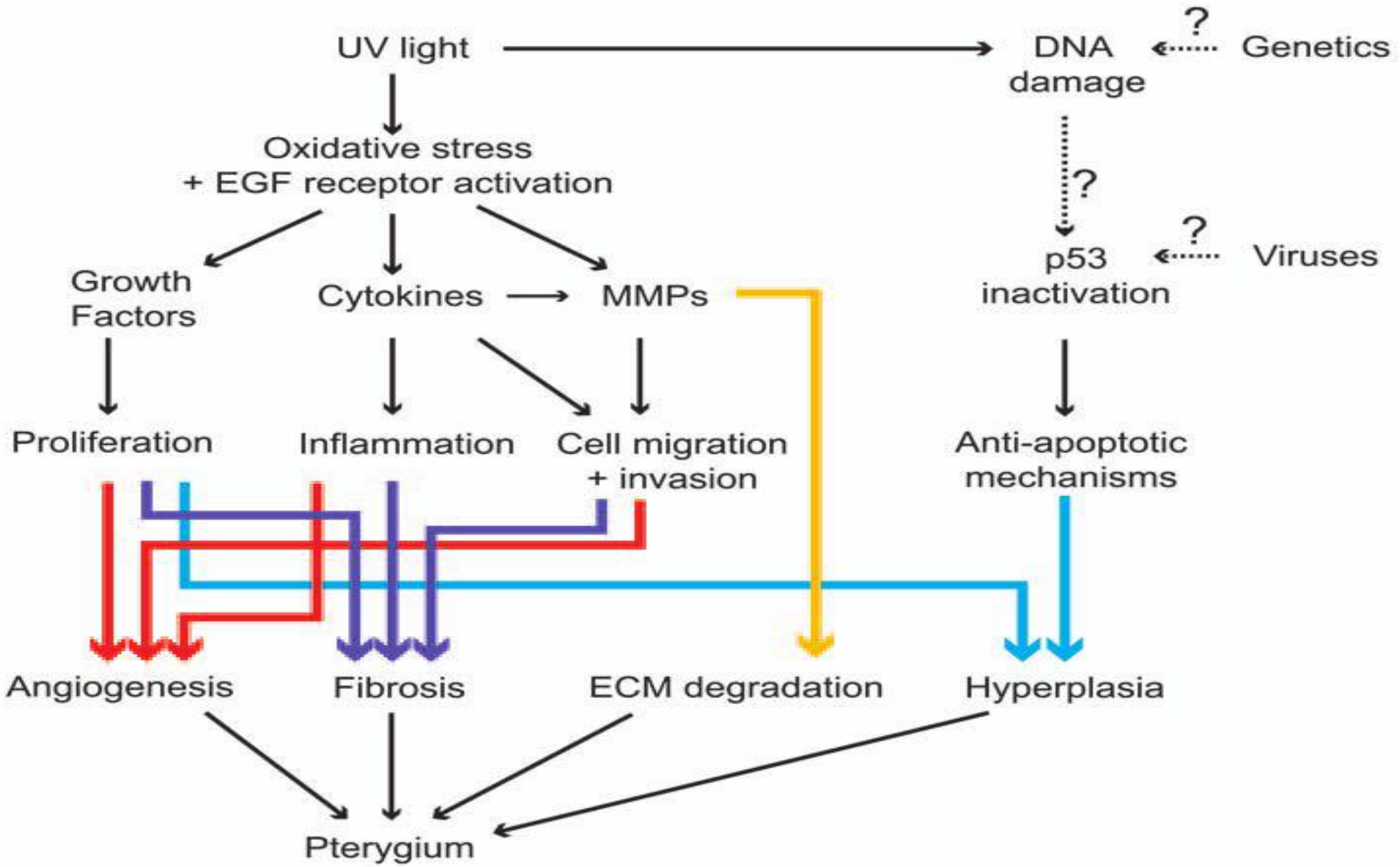
Fig. 20.6: Diagrammatic representation of Chao et al. staging system for pterygium according to the extent of the pterygium. (1—just crossing the limbus; 2—crossing the midway between limbus and pupillary margin; 3—crossing the pupillary margin)

ETIOPATHOGENESIS

- UV light
- Focal limbal cell deficiency
- Molecular alterations
- HPV
- Over expression of angiogenic factors and decrease in anti-angiogenic factors

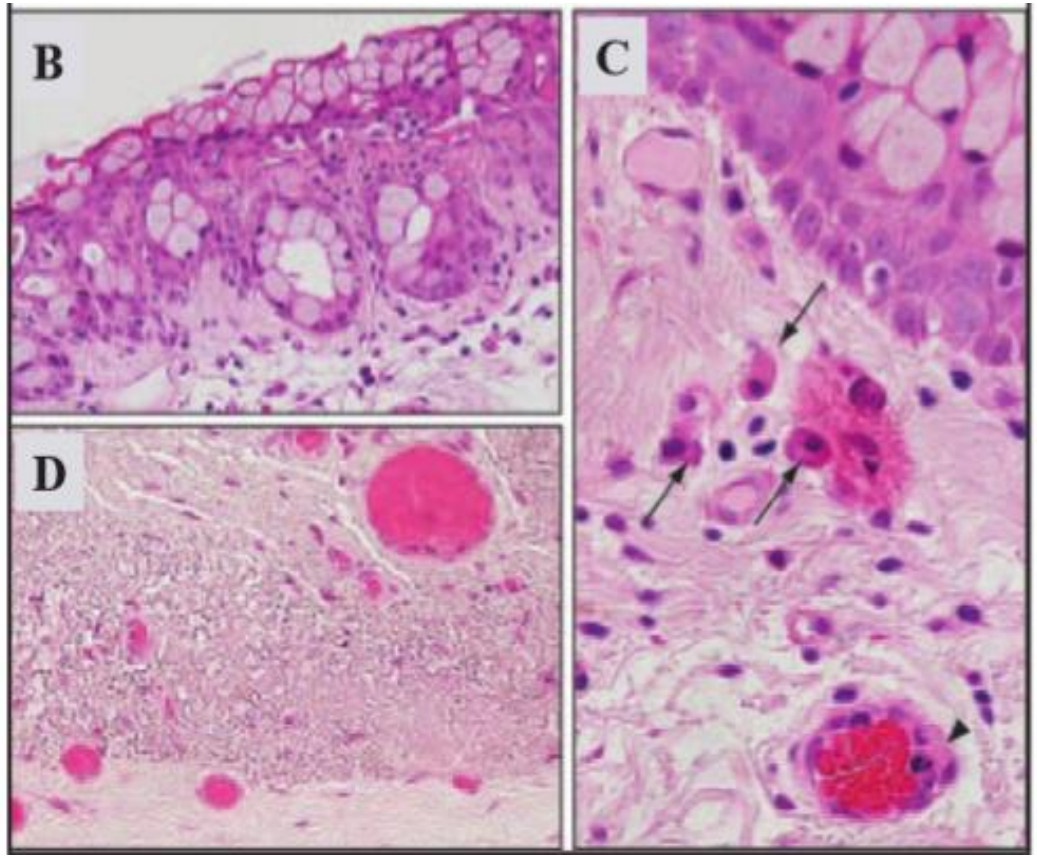
PROLIFERATION OF PTERYGIUM




UV light
DNA damage
?
Genetics
Oxidative stress
+ EGF receptor activation
?
p53 inactivation
?
Viruses
Growth Factors
Cytokines
MMPs
Proliferation
Inflammation
Cell migration + invasion
Anti-apoptotic mechanisms
Angiogenesis
Fibrosis
ECM degradation
Hyperplasia
Pterygium

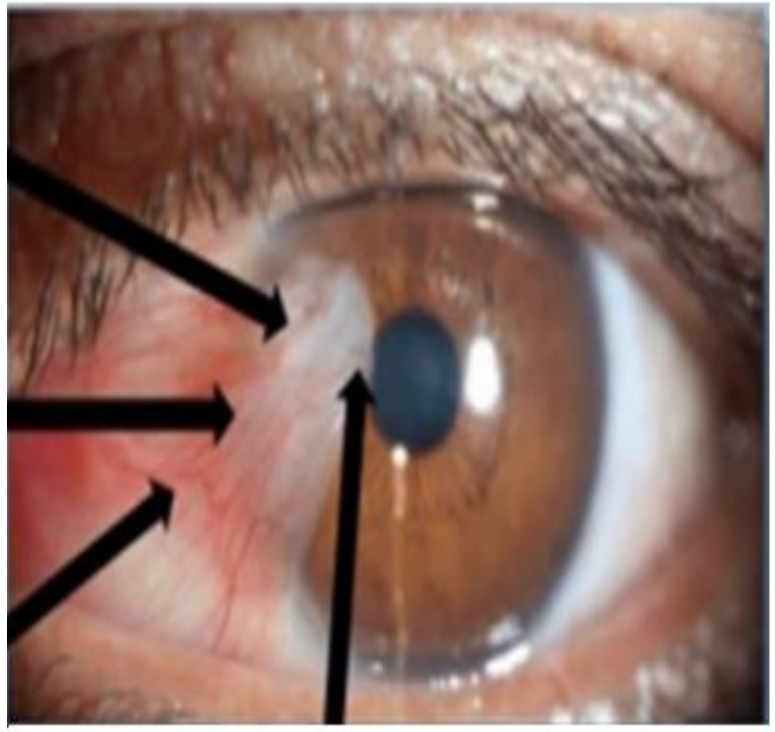
PATHOLOGY

- ELASTOTIC DEGENERATION of collagen
- Recurrent pterygium- no elastotic degeneration
- Pterygium epithelial cells are +ve for IHC for various MMPs



CLINICAL FEATURES

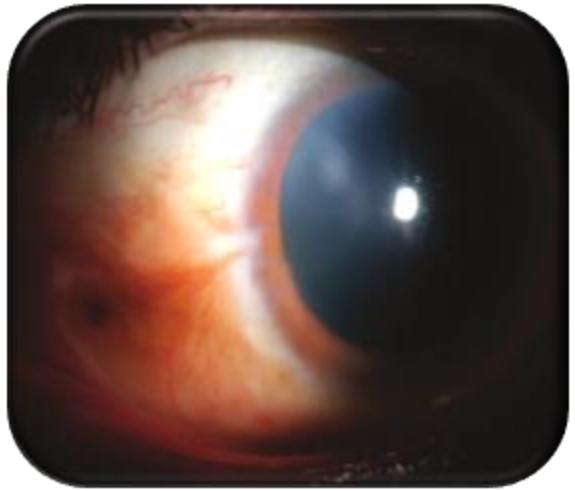
- PARTS –
  - Body- fleshy
  - Neck- limbus
  - Head- vascular part on cornea
  - Cap- fibrous non vascular leading edge
  - STOCKER’S LINE
  - FUCH’S PATHCHES*
- SYMPTOMS-
- DOV
- F/B sensation
- Irritation
- Cosmetic blemish



PREDICTORS OF RECURRENCE

- Morphology of the pterygium.
- Biomarker – Stromal cell Derived Factor 1 (SDF 1).
- Pre existing disfiguration of lacrimal caruncle.
- Concurrent ocular surface inflammation.
- Fibrogenic constitution (keloid tendency).

PROPOSED GRADING OF RECURRENT PTERYGIUM [1]

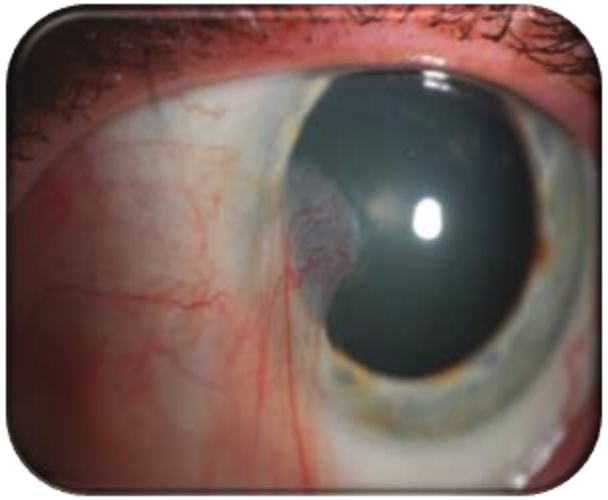


Grade 1 – Normal appearing operative site.

Grade 2 – Fine episcleral vessels.



Grade 3 – Additional fibrous tissues.



Grade 4 – Actual corneal recurrence.



PTERYGIUM AND ASTIGMATISM

- MC TYPE?
- MECHANISM
- Size – amount of astigmatism
  - >2mm
  - >5mm wide
  - >6.25mm sq area
  - Induces >2D astigmatism

PREVENTION



MANAGEMENT

- MEDICAL
  - Refractive correction
  - Topical lubricants
  - Topical corticosteroids – loteprednol etabonate 0.5%
  - UV filters in glasses

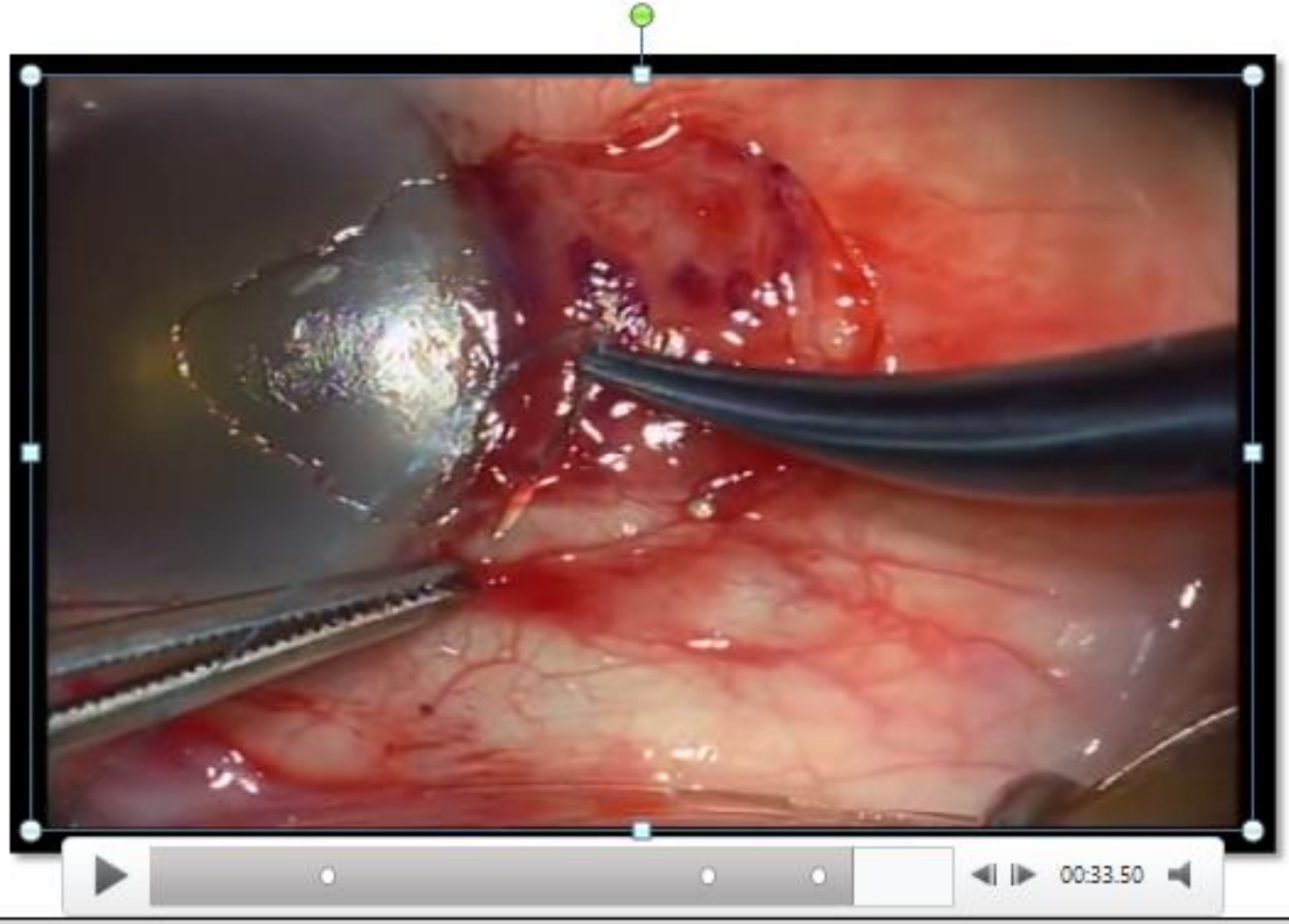
Investigation before surgical management*

SURGICAL

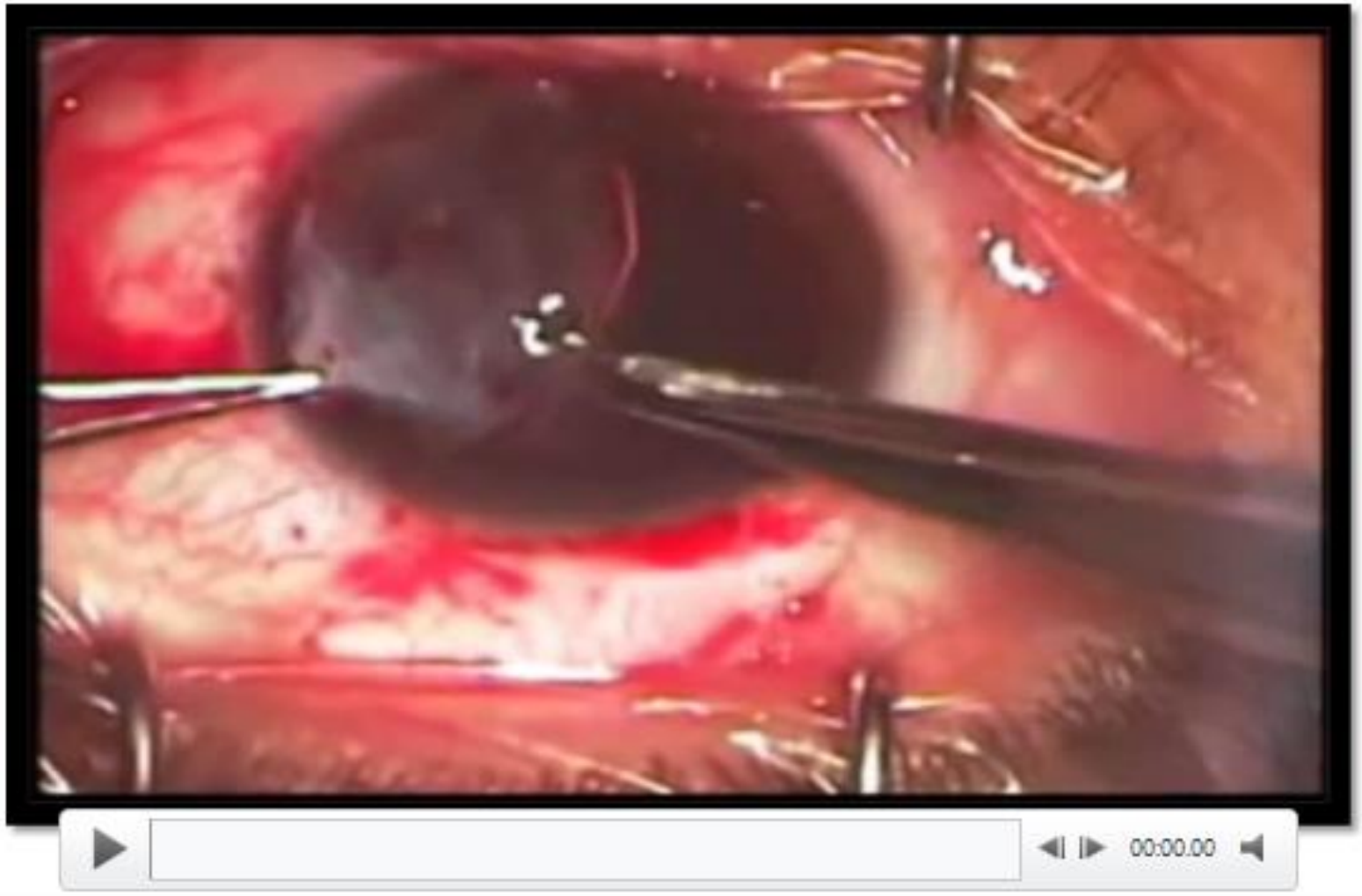
- INDICATIONS
- PRE- OP PREPARATION – r/o infections, plan when pt can avoid sunlight
- INTRA-OP CONSIDERATION-
  - Anesthesia
  - Topical dilating drops
- TECHNIQUES:
  - Avulsion
  - Superficial Keratectomy

- TYPES OF CLOSURES
  - BARE SCLERA- 30-80%
  - SIMPLE CONJUNCTIVAL CLOSURE- 29-89%
  - CONJUNCTIVAL LIMBAL AUTOGRAFT -2-40%
  - FREE CONJUNCTIVAL AUTOGRAFT- 2.6-39%
  - P.E.R.F.E.C.T. – 0.5%
  - CONJUNCTIVAL ROTATIONAL AUTOGRAFT
  - CULTIVATED CONJUNCTIVAL AUTOGRAFT- 23%
  - AMNIOTIC MEMBRANE GRAFT- 2.6-10.9%

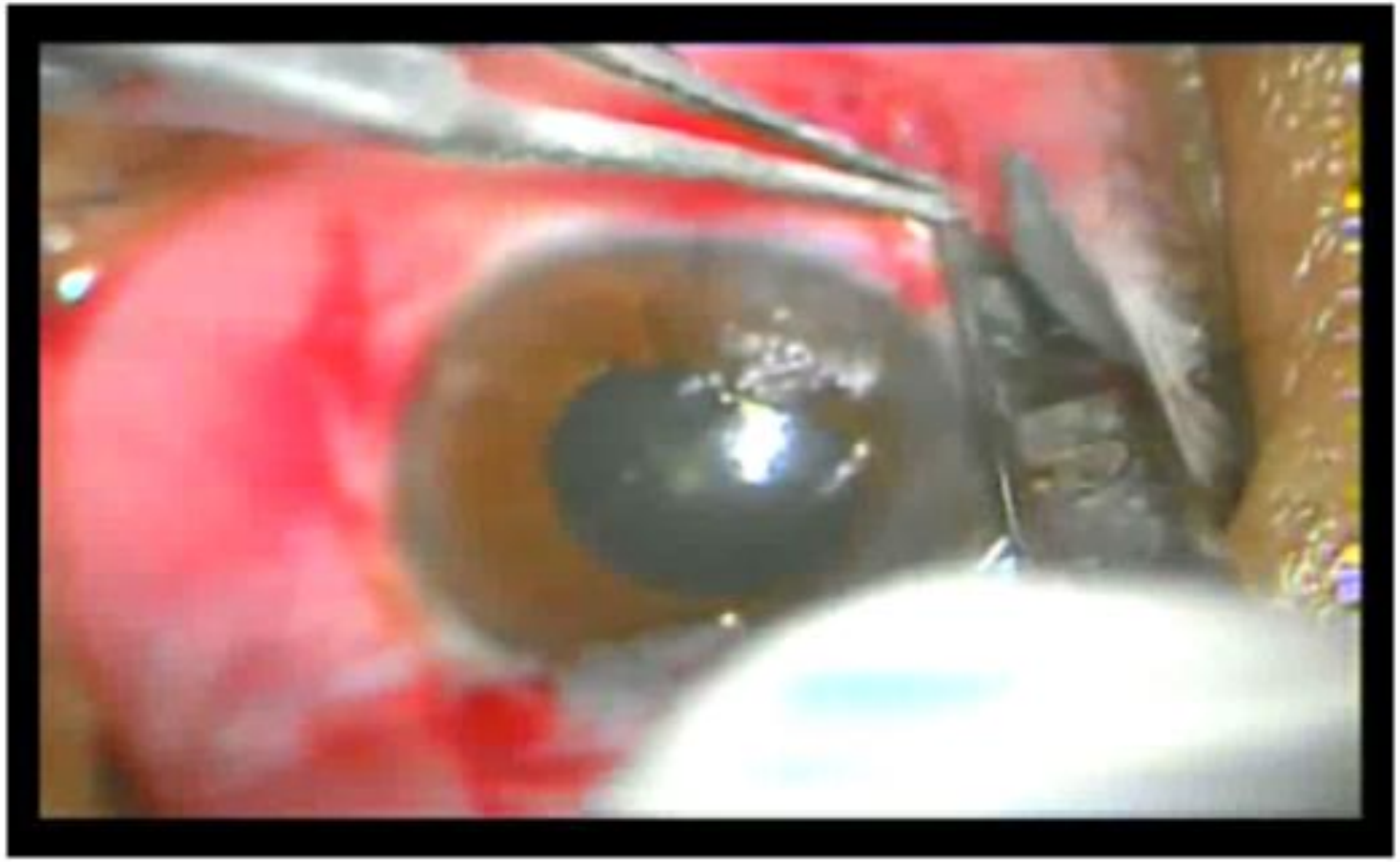
CONJUNCTIVAL AUTOGRAFT WITH SUTURES



FIBRIN GLUE

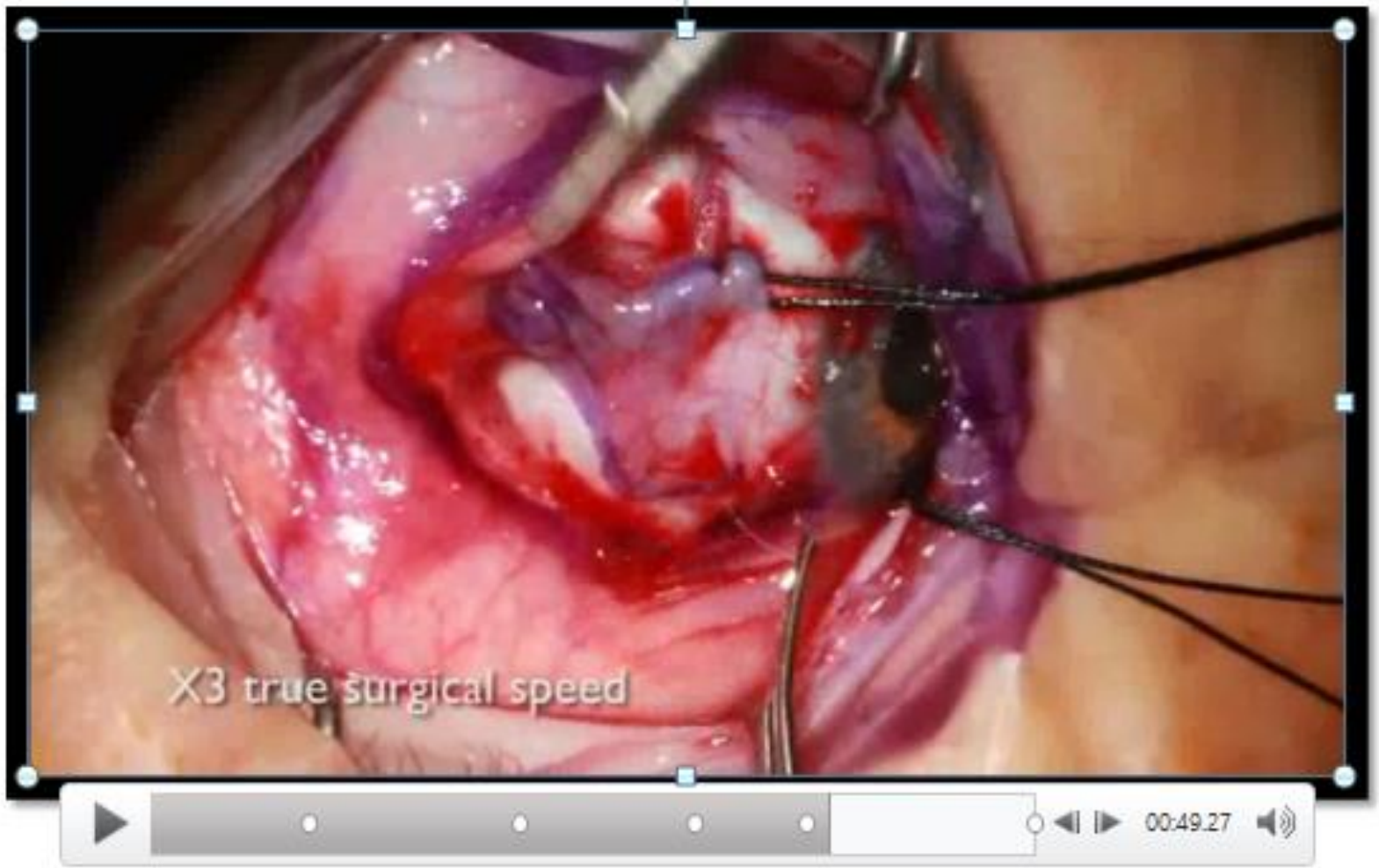


CUT AND PASTE –AUTOLOGOUS SERUM

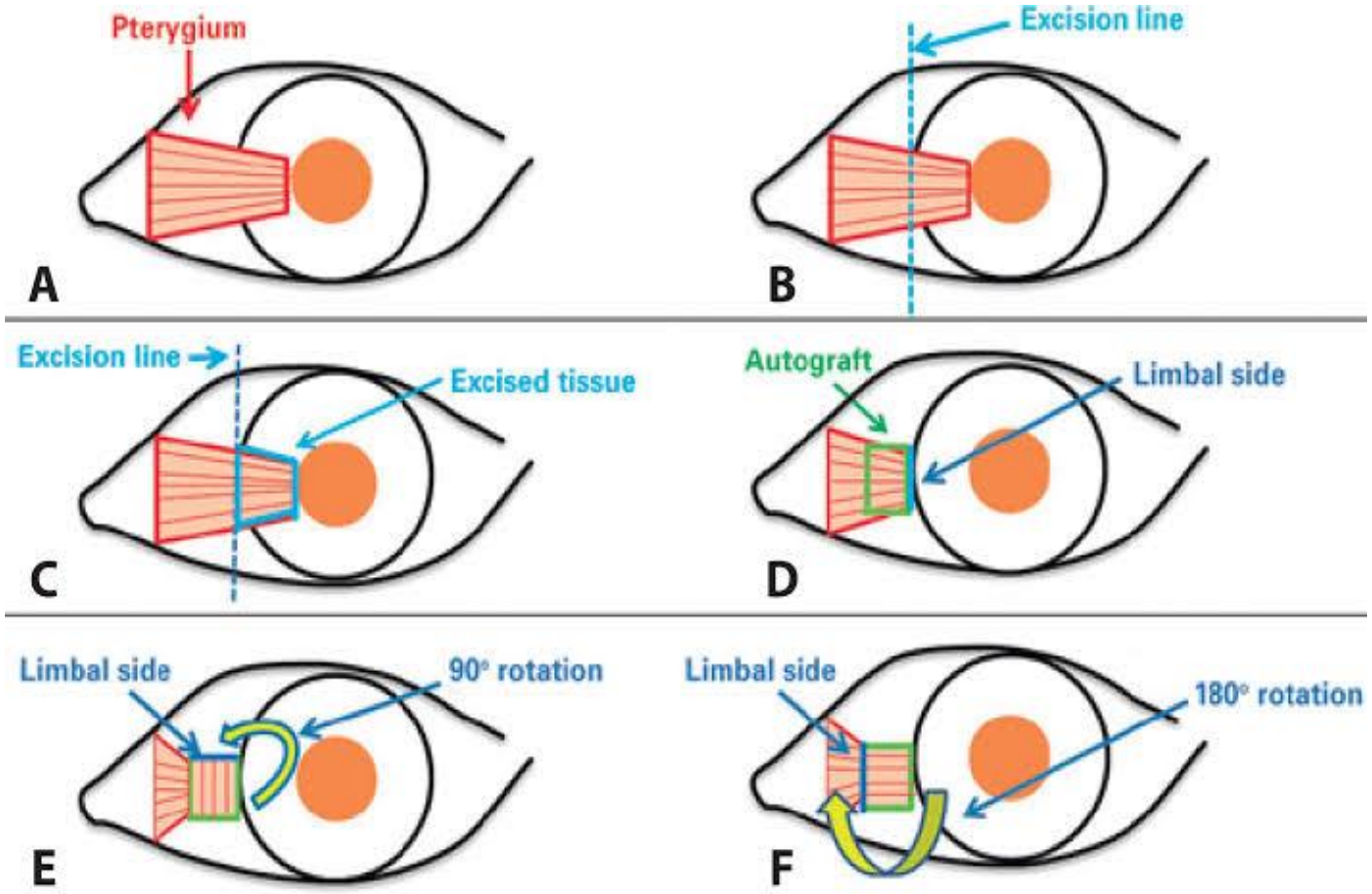


P.E.R.F.E.C.T SURGERY

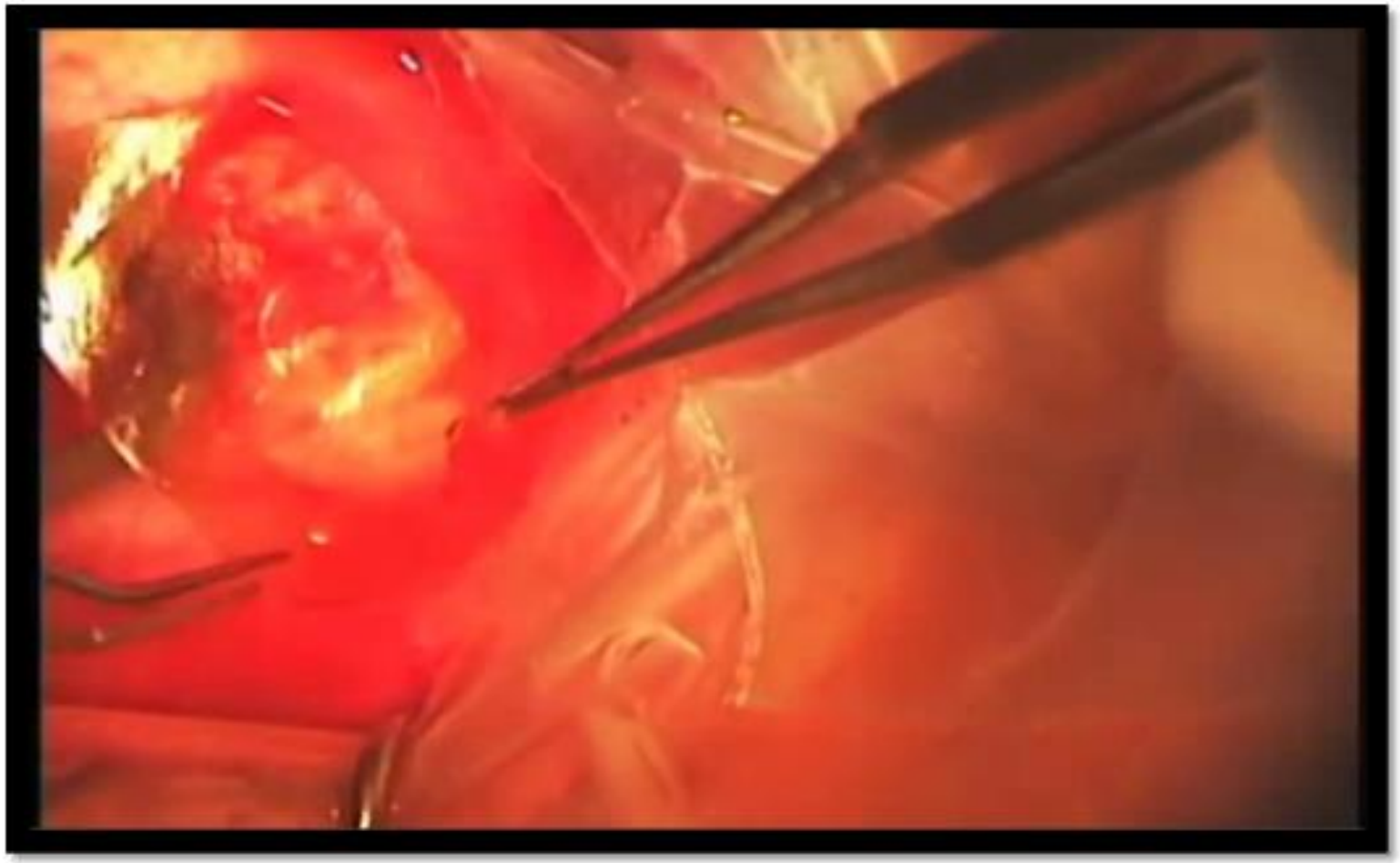
- Pterygium Extended Removal Followed by Extended Conjunctival Transplant.



CONJUNCTIVAL ROTATIONAL AUTOGRAFTS

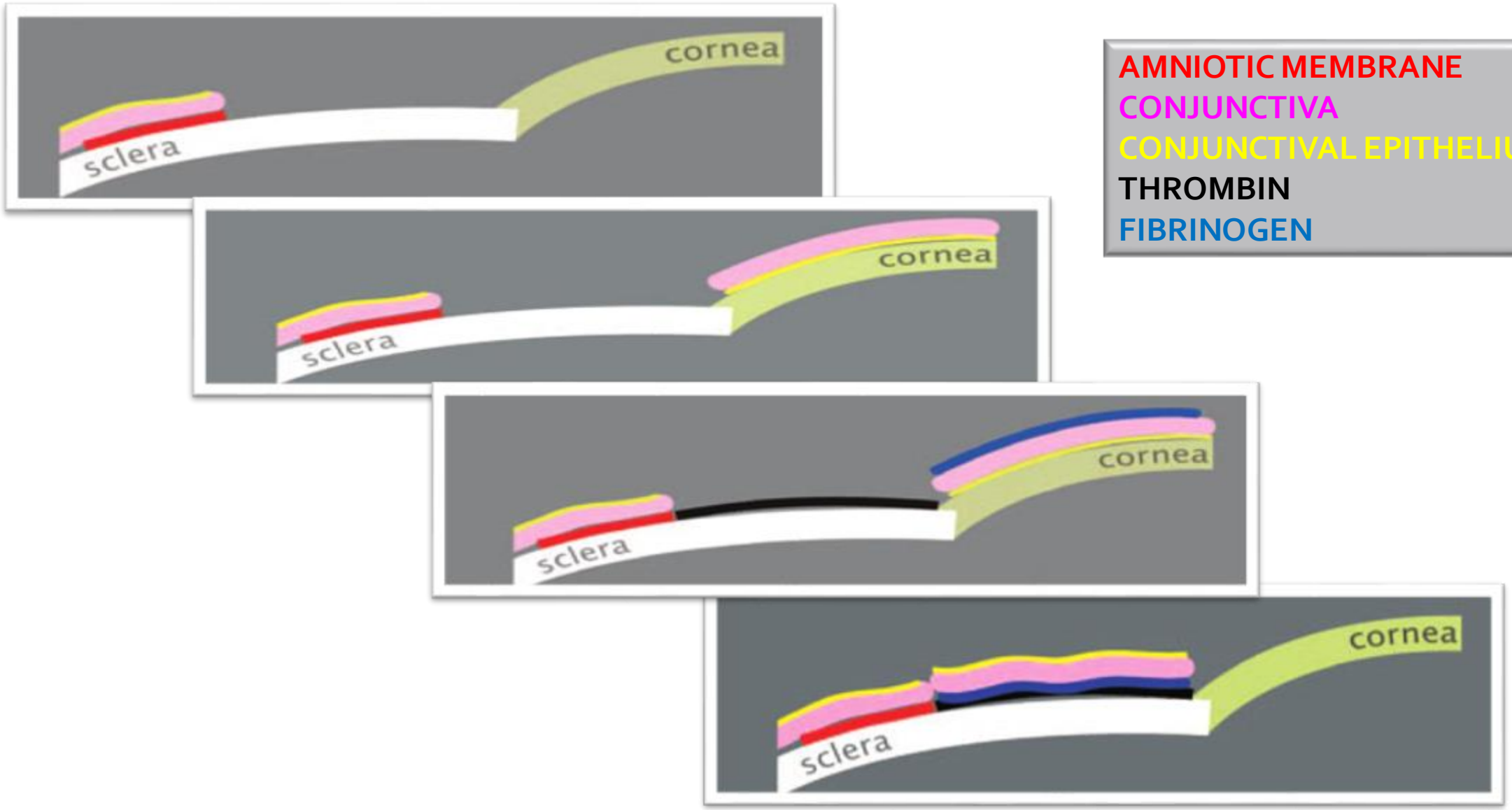


AMNIOTIC MEMBRANE GRAFT



COMBINATION OF APPROACHES

- CONJUNCTIVAL AUTO-GRAFTING + AMNIOTIC MEMBRANE GRAFTING –
- INDICATION - Recurrent pterygium with restriction of movements.
- AMNIOTIC MEMBRANE -
  - cover the muscle to prevent fibrosis and restriction of movements.
- CONJUNCTIVAL AUTOGRAFT -
  - close the defect from the pterygium surgery.



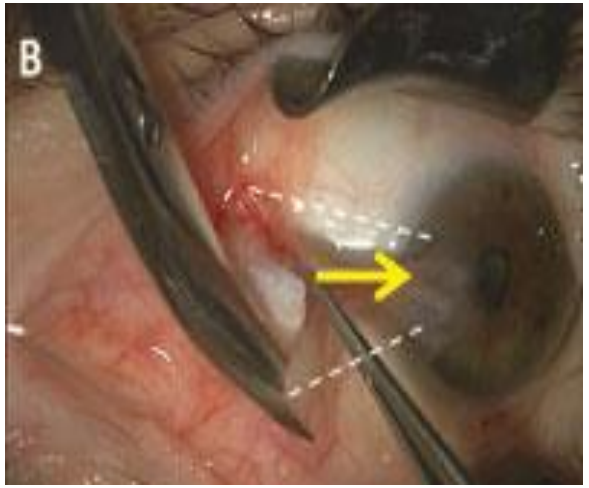
AMNIOTIC MEMBRANE
CONJUNCTIVA
CONJUNCTIVAL EPITHELIUM
THROMBIN
FIBRINOGEN

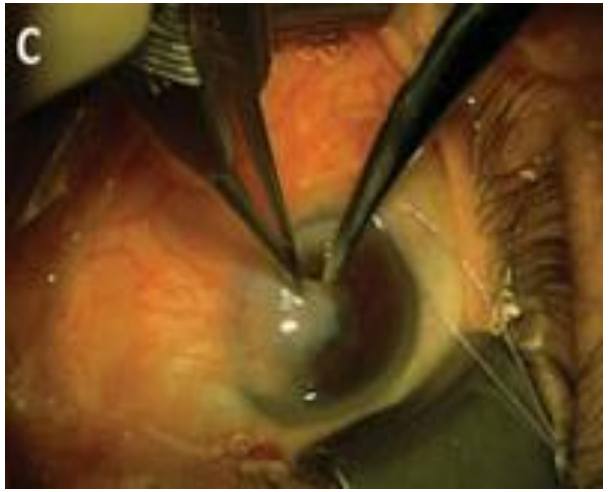
CORNEAL PROCEDURES

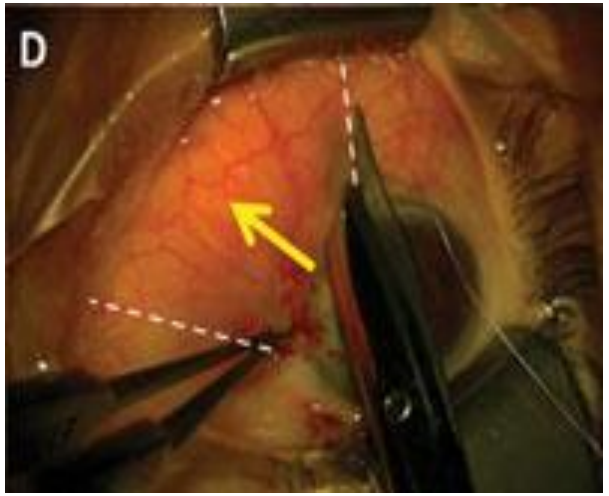
- Optical zone involvement requires corneal surgery.
  - Lamellar keratoplasty.
  - Phototherapeutic keratectomy (PTK).
  - Penetrating keratoplasty.
- Advantage:
  - Clears the optical axis.
  - Provides a barrier to reduce recurrence.

- For primary pterygium, incision anterior to the semilunar fold (A) to resect the fibrovascular tissue, including the Tenon, from the cornea (B, arrow).
- For recurrent pterygium, incision from the head by lifting the remaining conjunctiva from the cornea (C) and being recessed to the fornix without resection (D, arrow).









ADJUVANT TREATMENT

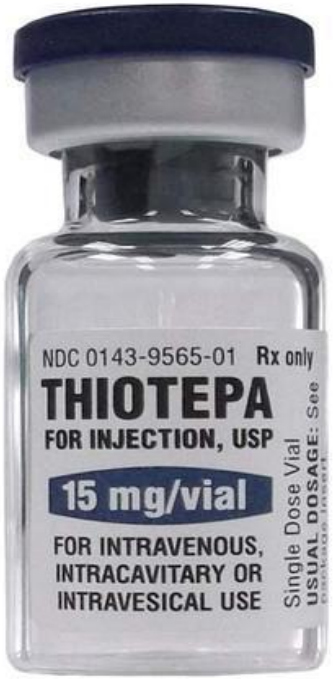
- **CYCLOSPORINE –**



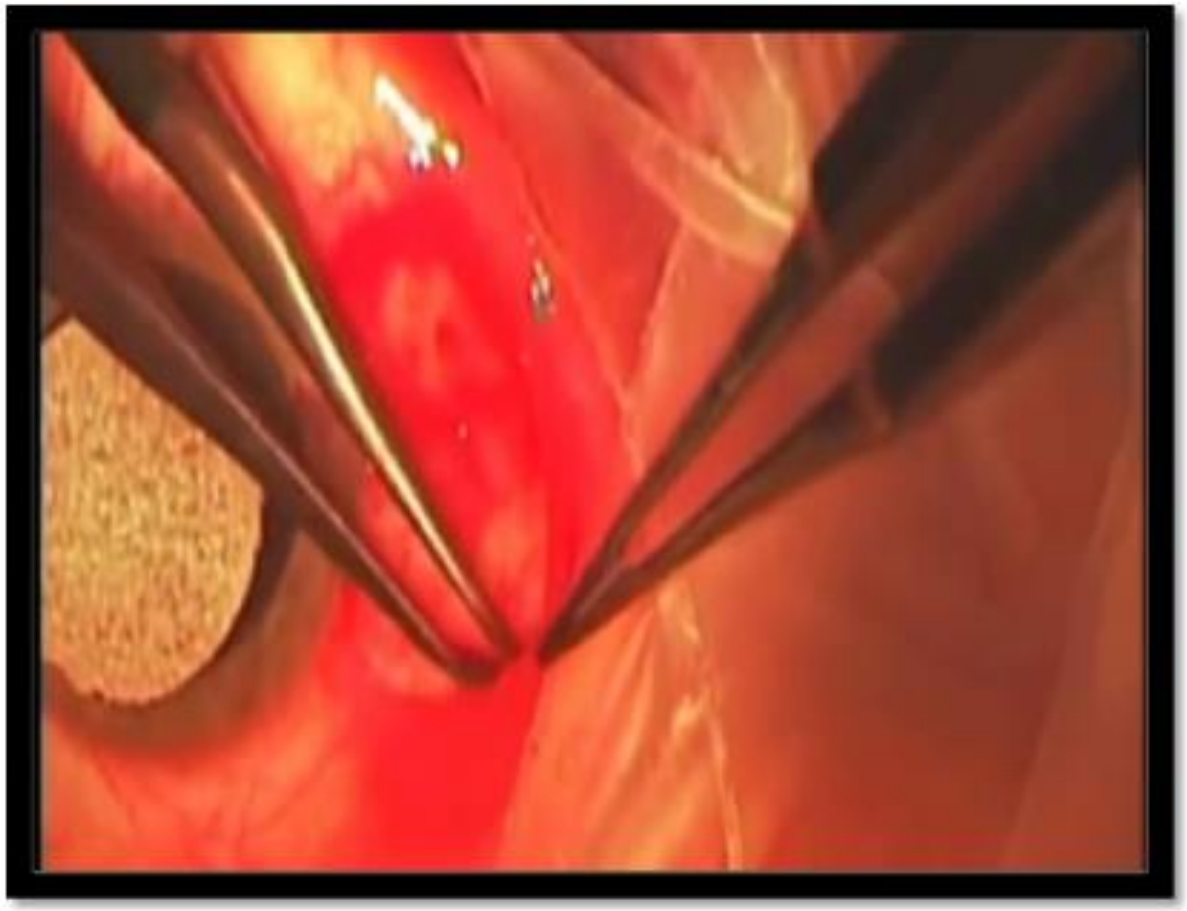
- **5 FLUOROURACIL –**

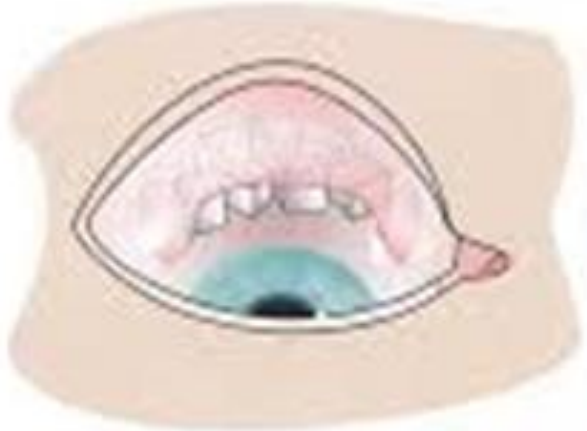


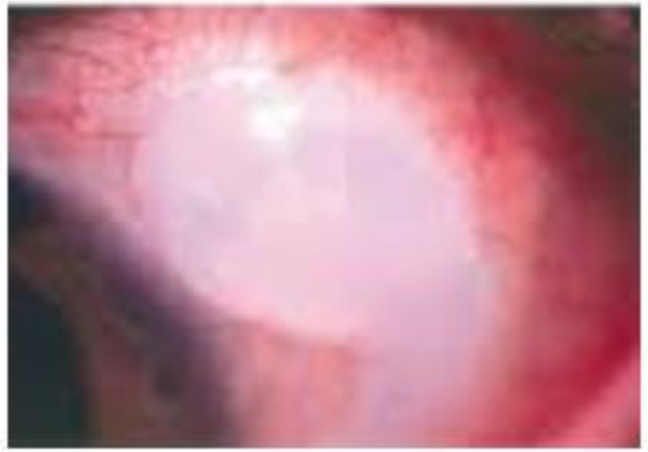
- **THIOTEPA-**



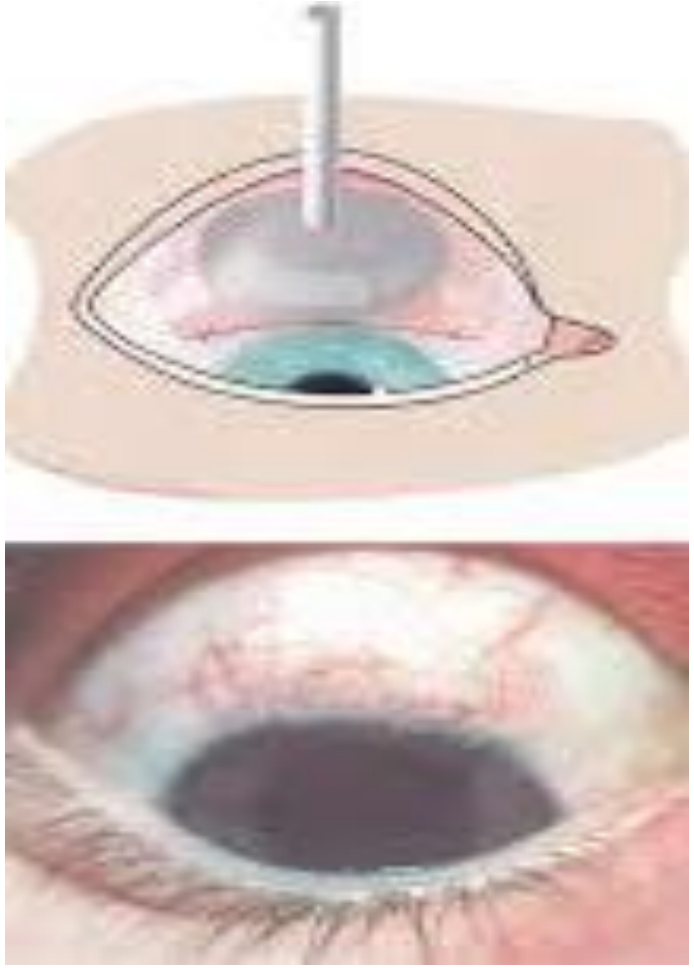
- **MITOMYCIN C –**







- **BETA RADIATION-**



- **BEVACIZUMAB (AVASTIN)-**



- **OTHERS-**
- Multi microporous expanded Poly Tetra Fluoro Ethylene (e-PTFE).
- Tacrolimus.
- Interferon alpha 2 b.
- Doxorubicin and Daunorubicin.

- POST-OP CARE
- COMPLICATIONS-
  - INTRA-OP
    - Globe perforation
    - Inverted graft
    - Bleeding
    - Button holing
    - EOM damage
  - IMMEDIATE
    - Chemosis and Hg
    - Hematoma
    - Epithelial defects
    - Loose sutures
    - Infection
  - LATE
    - Diplopia
    - Strabismus
    - Pyogenic granuloma
    - Tenons cyst

FUTURE PROSPECTS

- ANECORTAVE ACETATE-

  Inhibits angiogenesis.

  Topical 1% - inhibitory effect on pterygium regrowth.

- CRISPR-Cas9 system 9 (clustered regularly interspaced short palindromic repeats) or “gene scissors” .

Case

- A 73 year old gentleman
- Farmer
- No comorbidities
- C/o diminution of vision in both eyes (RE>LE) since 5 years
- C/o foreign body sensation and a pink fleshy mass in the RE since 6 months



PTERYGIUM AND CATARACT

- IMPACT ON VISUAL PARAMETERS
- SURGERY
  - SIMULTANEOUS OR SEPARATE
  - TIMING – 1-3 MONTHS

REFERENCES

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- Kim KW, Kim JC. Current approaches and future directions in the management of pterygium. Int J Ophthalmol. 2018;11(5):709-711.
- Bahuva A, Rao SKCurrent Concepts in Management of Pterygium.DJO 2014;25:78-84.
- Stephen C. Kaufman et al. Options and Adjuvants in Surgery for Pterygium: A Report by the American Academy of Ophthalmology.
  January 2013, Vol 120, 201-208.
- Peyman textbook of ophthalmology

THANK YOU